

Introduction

- As technology has innovated, global consumption and exposure to screens has increased exponentially. With that, a larger population of our youth are using this technology.
- But while convenient and beneficial, overexposure to blue light from technology has detrimental effects including circadian rhythm disruption (Zhao et al., 2018), vision problems (Arthaut et al., 2017), lower lifespan and cognitive function (Nash et al., 2019).
- Drosophila melanogaster* have been well established as great model organisms in behavioral studies due to their short life cycles (Jennings, 2010), and similarity to humans including biological mechanisms and behaviors (Jennings, 2011; Tolwinski, 2017).

Figure 1: Wild-type *D. melanogaster*



Purpose

- Previous literature solely studies the impacts of blue light on a specified stage of growth (e.g. adulthood, larvae, etc.). This study aims to study the effects blue light has in adulthood when there is an early onset of overexposure.

Research Question:

- Does over-exposure to blue light to *Drosophila* during their larval stage have a negative impact on behavior and physical characteristics when they reach adulthood?

Hypothesis:

- If *Drosophila* are exposed to blue light during larval development, then adult flies will exhibit decreased length and altered courtship behaviors compared to controls because blue light disrupts neurodevelopmental and circadian processes.
- IV: Light cycle exposed
- DV: Growth/Development (length), Behavior (aggression + courtship assays)
- Control: 12-hour day-night cycle

Effects of Blue Light on the Growth Progression of *Drosophila Melanogaster*

Experimental Design

Figure 2: Methodology for behavioral assays (Yadav et al., 2025)

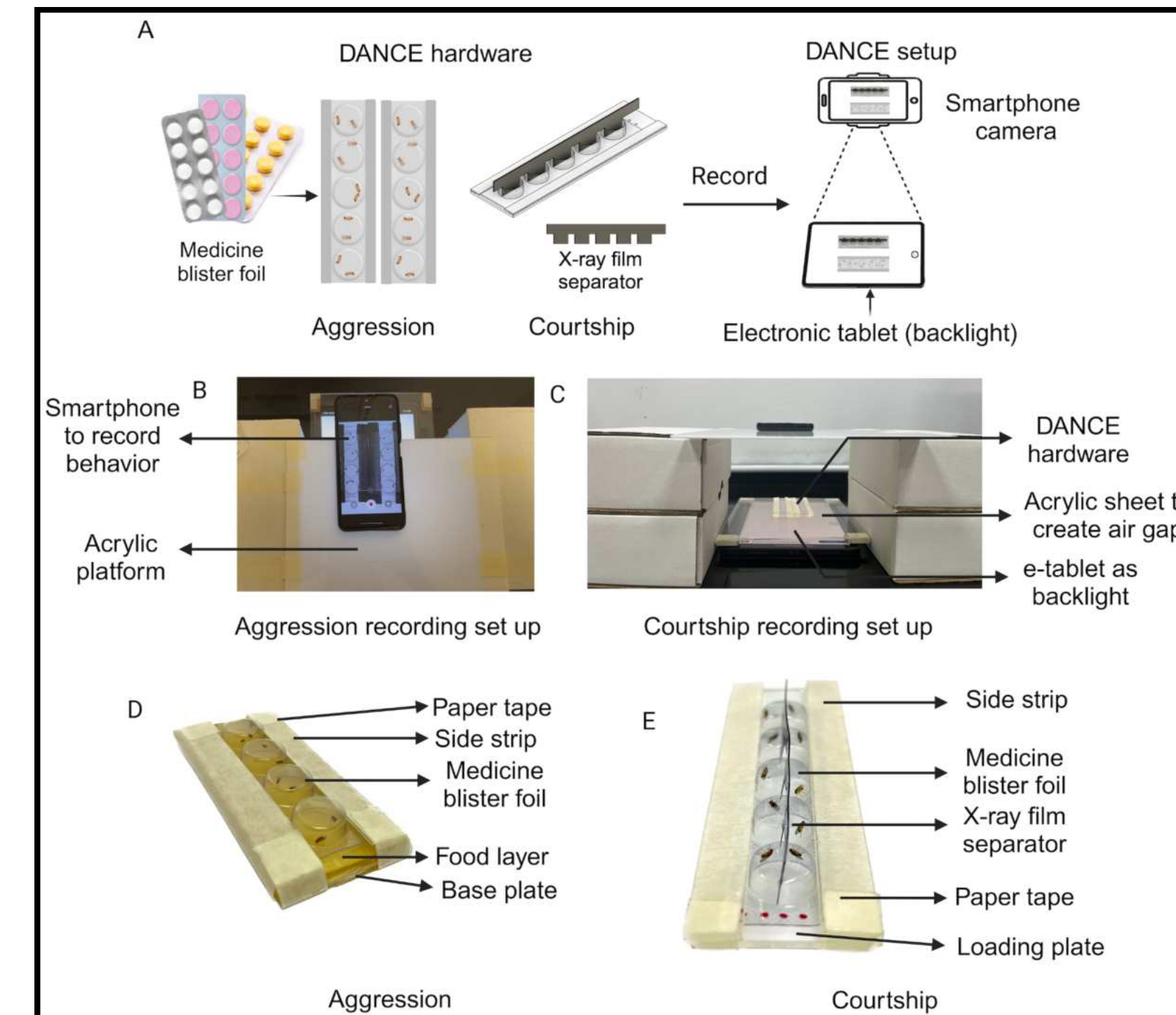
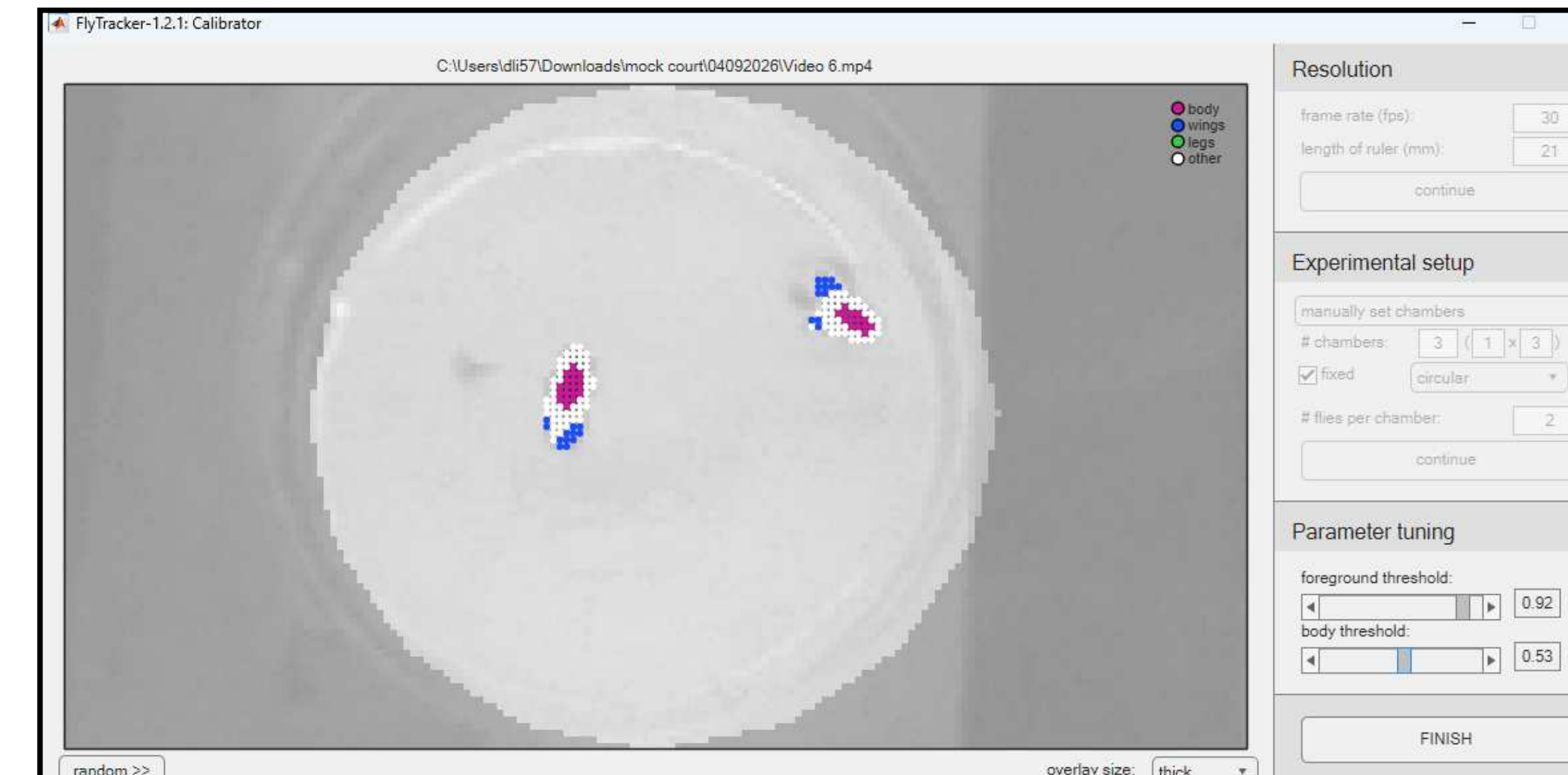


Figure 3: DANCE model analyzing courtship assay within MATLAB



Results

Figure 4: Length of *Drosophila* for Each Group

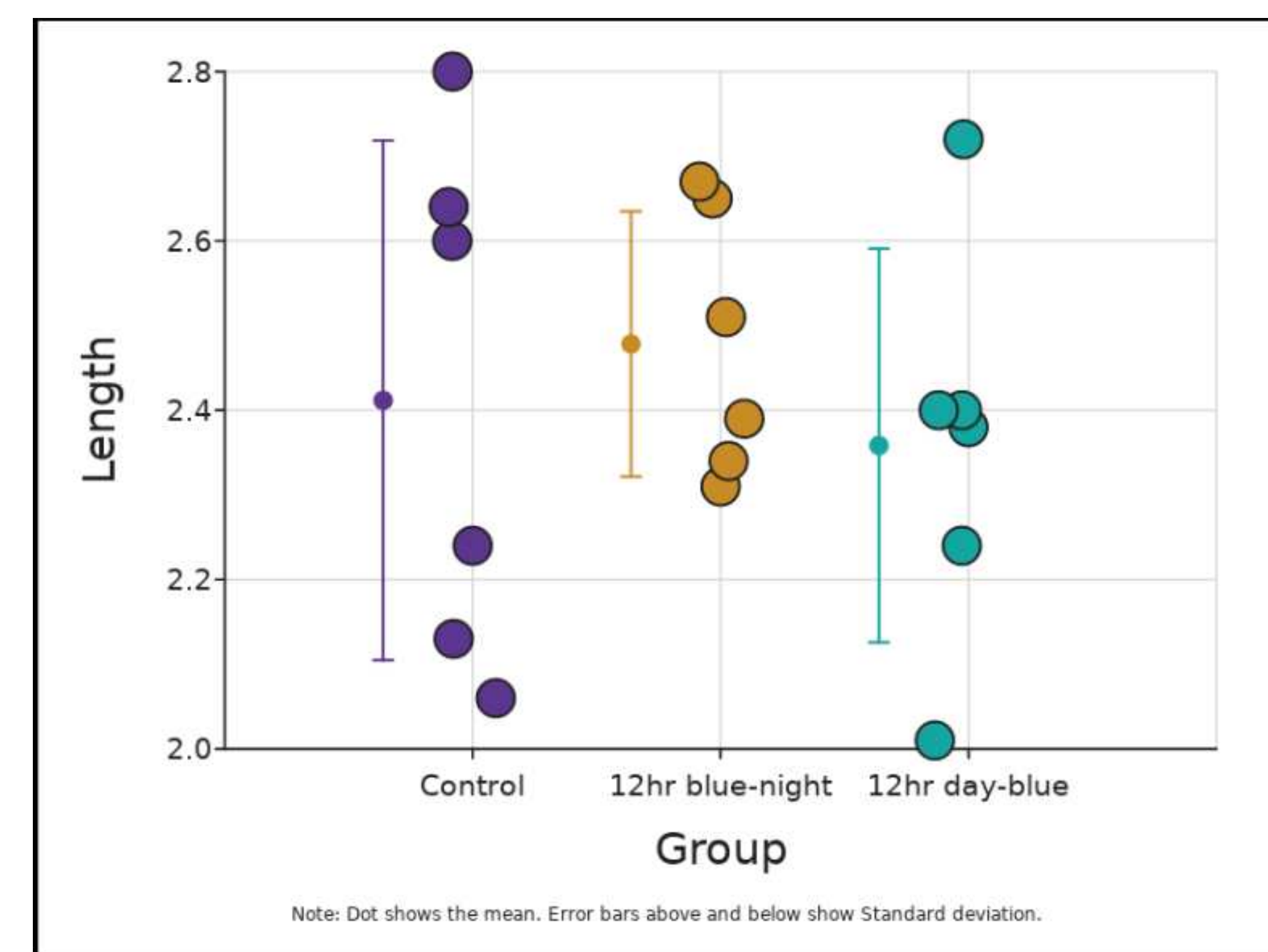


Figure 5: Quantity of Attempted Copulation Actions for Each Group

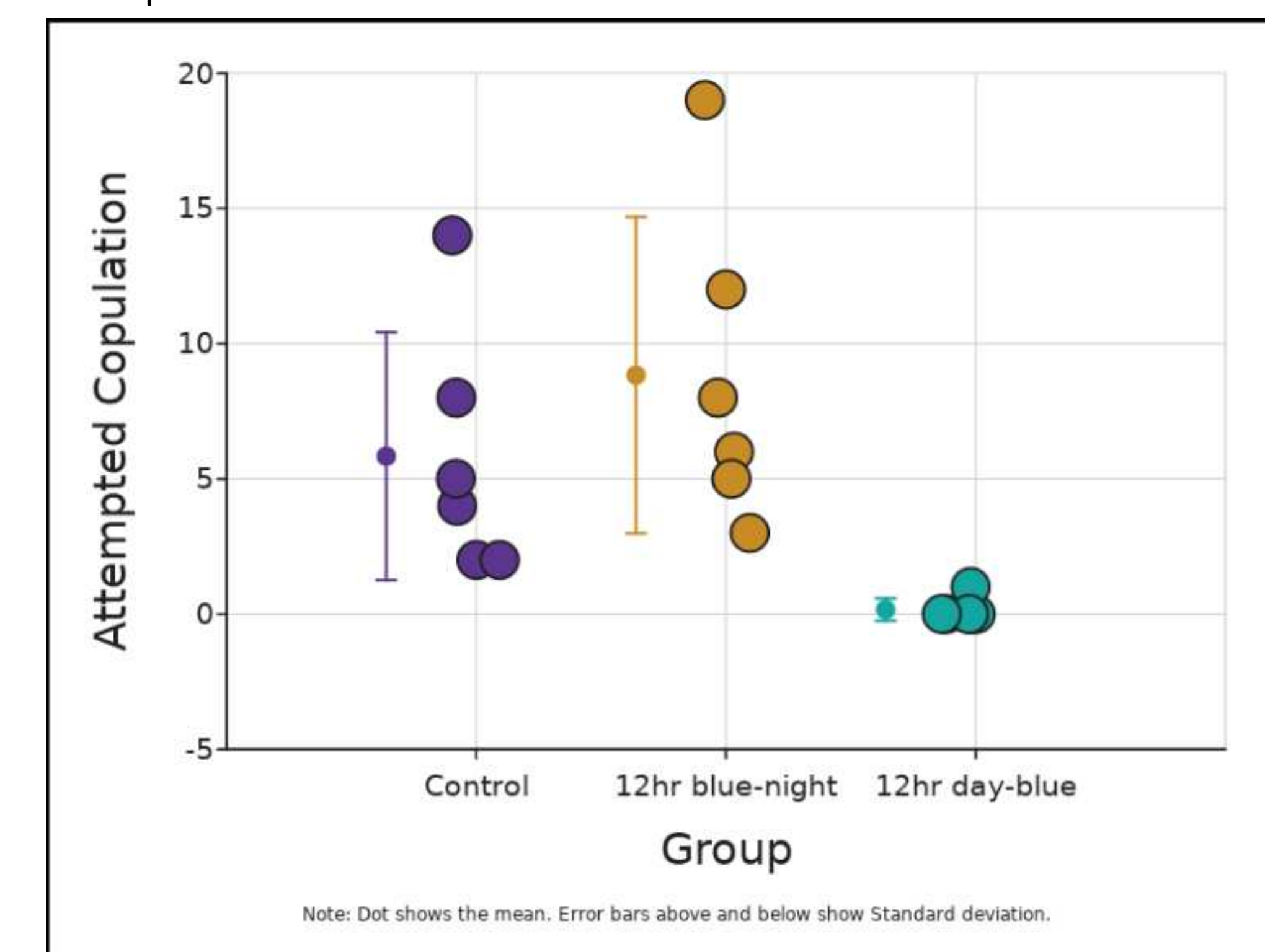
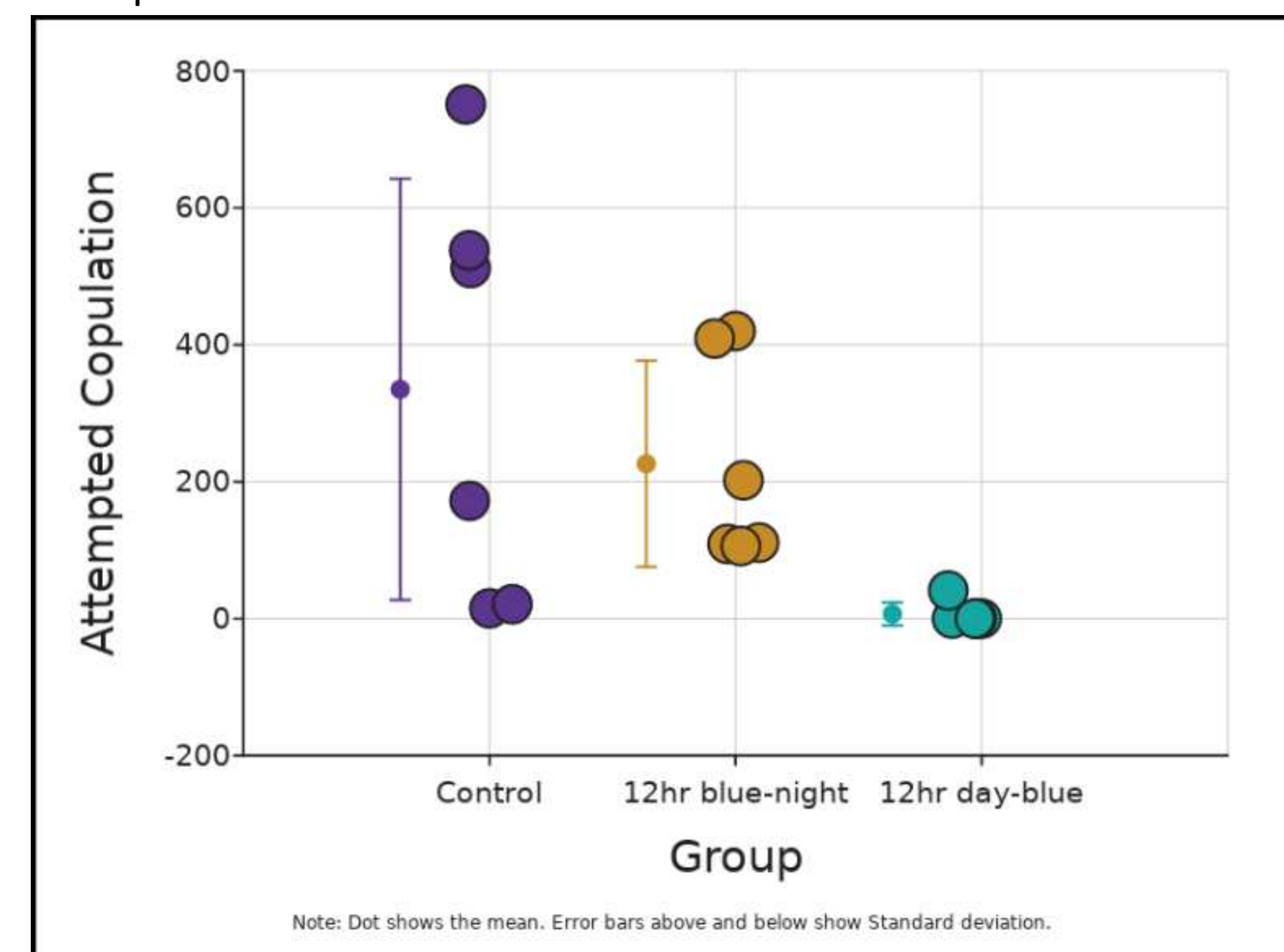


Figure 6: Duration of Attempted Copulation Events for Each Group



Observed Trends

- No significant change in fly length between each group
- Significant drop-off in attempted copulation events when flies were exposed to the 12hr day-blue light cycle
- Significant drop-off in average time spent in attempted copulation events when flies were exposed to the 12hr day-blue light cycle
- Control groups showed the largest variance indicating that more trials are needed in order to discount the possibility of skews/outliers

Statistics

- Kruskal-Wallis test was used as the assumptions of ANOVA could not be met
- $\alpha = 0.05$
- Growth/Development: Length
 - $p = 0.74$
 - No evidence that the groups might be different
- Quantity of Attempted Copulation Actions
 - $p < 0.01$
 - Strong confidence that the groups are different
- Duration of Attempted Copulation Events
 - $p < 0.01$
 - Strong confidence that the groups are different

Conclusion

- The statistical analysis finds there is **no indication** that blue light has any significant impact **on stunting the physical growth** of *Drosophila*
- However, there is **evidence** that **blue light** does **alter the courtship behaviors** of *Drosophila*
 - 12hr day-blue light cycle** experiences **fewer frequency** of copulation events and **lower duration** of each event
- Results provide further insight into the **negative effects of blue light** on behavior mechanisms in adulthood
- Future research could further reveal potential generational impacts blue light stress has on offspring health

References

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